

Lecture 5: Estimation of the Multiple Linear Regression Model

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1 Functional Relationships (Review from Lecture 2)

We consider that we have a $k + 1$ vector $(X_1, \dots, X_k, Y)$ where X_i 's are the explanatory variables, these variables include

- Variables of Economic Interest (We are interested in the C.P. effect)
- Control variables (They do affect Y, but they do not have economic interest for us).

We may assume that there is a joint pdf

$$f(x_1, \dots, x_k, y) = f(\mathbf{x}, y)$$

such that the conditional expectation is defined as:

$$E[Y|X_1 = x_1, \dots, X_k = x_k] = \int_{-\infty}^{\infty} y f(y|x_1, \dots, x_k) dy = h(\mathbf{x}) = E[Y|\mathbf{X} = \mathbf{x}]$$

where the conditional pdf of Y is: $f(y|x_1, \dots, x_k) = \frac{f(x_1, \dots, x_k, y)}{f_{x_1, \dots, x_k}(x_1, \dots, x_k)}$

Remark: The CEF $h(\mathbf{x}) = E[Y|\mathbf{X} = \mathbf{x}]$ may or may not be a linear function of $\mathbf{X}$

(Example in Lecture 2 is non-linear, as well as the example in Midterm 1, exercise 1)

The Ceteris Paribus effect of X_i on Y is $\frac{\partial h(\mathbf{x})}{\partial x_i}$

In empirical applications, the conditional pdf is not known, the CEF cannot be calculated. In this case we will estimate the BLP:

$$E^*(Y|X_1, \dots, X_k) = \beta_0 + \beta_1 X_1 + \dots + \beta_k X_k$$

which was defined in Theorem 2.10. In this case the Ceteris Paribus effect of X_i on Y is given by:

$$\frac{\partial E^*(Y|X_1, \dots, X_k)}{\partial X_i} = \beta_i$$

(Provided X_i appears only as a linear term, if powers or other functions appear you need to account for that)

Example from Lecture 2 $E^*(Y|X, X^2) = \beta_0 + \beta_1 X + \beta_2 X^2$

The CP effect of X on Y : $\frac{\partial E^*(Y|X, X^2)}{\partial X} = \beta_1 + 2\beta_2 X$

Recall that $\beta_p = (\beta_1, \beta_2, \dots, \beta_k)^T$ is the "vector of slopes" of the BLP, then from Theorem 2.10

$$\beta_p = [V(\mathbf{X})]^{-1} Cov(\mathbf{X}, Y)$$

$V(\mathbf{X})$ is the variance-covariance matrix of $\mathbf{X}$

$Cov(\mathbf{X}, Y)$ is the vector of covariances of variables in $\mathbf{X}$ with Y .

From Theorem 2.10, $\beta_0 = E[Y] - \beta_1 E[X_1] - \dots - \beta_k E[X_k]$

In section 3 of this lecture we will estimate the whole vector

$$\beta = (\beta_0, \beta_1, \dots, \beta_k)^T$$

2 Interpretation of the Coefficients

$$E^*(Y|X) = \beta_0 + \beta_1 X \quad (1)$$

$$\Rightarrow \frac{\partial E^*(Y|X)}{\partial X} = \beta_1 \text{ The Ceteris Paribus effect of } X_1 \text{ on } Y \quad (2)$$

The Ceteris Paribus effect of X_1 on Y

This generalizes to multiple explanatory variables:

$$E^*(Y|X_1, X_2, \dots, X_k) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k \quad (3)$$

$$\frac{\partial E^*(Y|X_1, X_2, \dots, X_k)}{\partial X_j} = \beta_j \text{ for } j = 1, \dots, k \text{ The Ceteris Paribus effect of } X_j \text{ on } Y \quad (4)$$

For example, if Y savings and X is income/capital (in \$USD)

→ suppose the regression coefficient of X is 0.23: the average savings increases 23 cents for each dollar that the per capita income increases. A confidence interval say (0.20, 0.26) at 95% would be interpreted the same.

Sometimes, in order to improve how you communicate the estimated coefficients, i.e., the estimated ceteris paribus effects, it might be better to transform the model before estimating it.

→ in the previous example, instead of income and savings be in dollars, you may estimate it in thousands of dollars: $\tilde{Y} = \frac{Y}{1000}$, $\tilde{X} = \frac{X}{1000}$

Sometimes the units do not ?? nicely...

→ Say Y is in millions of dollars (5)

X is in cubic tons (6)

⇒ β_1 would be in millions of \$USD/cubic tons

Consider the arithmetic return, which is the decimal change between two quantities y_a and y_b

$$h = \frac{y_b - y_a}{y_a}$$

- h is unitless
- if you multiply h by 100% then $h(100\%)$ is in percentage points

A useful transformation is the natural logarithm of the variable. In order to relate h to the log-transformation:

$$y_b = y_a + y_a h = (1 + h)y_a$$

natural log on both sides yields:

$$\begin{aligned} \log y_b &= \log(1 + h) + \log y_a \\ \Rightarrow \log y_b - \log y_a &= \underline{\log(1 + h)} \text{ this side is the log return} \end{aligned}$$

Consider the Taylor series of $\log(1 + h)$ around $h = 0$,
by Taylor's theorem:

$$\log(1 + h) = \log(1) + \frac{\log^{(1)}(1)}{1!}h + \frac{\log^{(2)}(1)}{2!}h^2 + \frac{\log^{(3)}(1)}{3!}h^3 + \dots = \sum_{n=0}^{\infty} \frac{(-1)^n}{n+1}h^{n+1}$$

A first-order approximation is: $\log(1 + h) = \log(1) + \frac{\log^{(1)}(1)}{1!}h + \underline{R_2(h)}$, where
 $R_2(h) = \xi_2(h)h^2$, where $\lim_{h \rightarrow 0} \xi_2(h) = 0$

All of this in practical terms:

$$\begin{aligned}\log(1 + h) &\approx \log(1) + \frac{\log^{(1)}(1)}{1!}h \\ &= 0 + h \\ &= h\end{aligned}$$

$\therefore \log(1 + h) \approx h$ when h is small, i.e. $\log(1 + h)$ approximates the arithmetic return

$\Rightarrow \log(y_b) - \log(y_a) \approx h$ when h is small.

Summarizing the previous discussion:

$$\Delta \log(Y) \approx h \text{ **OR** } \Delta \log(Y)(100\%) \approx h(100\%)$$

whenever h is small.

Suppose we transform both X and Y using the natural log.

$$\beta_1 = \frac{\partial \log(Y)}{\partial \log(X)} = \frac{\Delta \log(Y)}{\Delta \log(X)} = \frac{\frac{\Delta Y}{Y}}{\frac{\Delta X}{X}} = \frac{\Delta \%in Y}{\Delta \%in X} = \text{elasticity of } Y \text{ w.r.t. } X$$

$$\rightarrow \Delta \log X = 0.01 \rightarrow \Delta \log(X)(100\%) = 1\%$$

$\Rightarrow \Delta \log(Y)(100\%) = \beta_1 \Delta \log(X)(100\%) = \beta(1\%) \therefore \beta_1$ is interpreted as the percent of Y due to a percent increase in X .

Example: $\beta_1 = 0.35$ is interpreted as a 1% increase in X produces a 0.35% increase in Y

Example $\beta_1 = 1.32$, a 1% increase in X produces an increase of 132% in Y .

$\therefore$ when both X and Y are in logs, the coefficient on X is **interpreted as an elasticity**. Y is the only variable in logs, then the coefficient on X has the interpretation of

$$\beta_1 = \frac{\Delta \log(Y)}{\Delta X} \rightarrow \log \text{ return of } Y \text{ from a unit increase in } X$$

An interpretation in terms of percentage points is:

$$\Delta X = 1 \Rightarrow \Delta \log(Y)(100\%) = \beta(100\%)$$

Ex. $\beta = 0.028$ then $\beta(100\%) = 2.8\% \Rightarrow 1$ unit in X yields an increase of Y by 2.8%

This last one is the empirical counterpart of

$$\frac{\partial \log Y}{\partial X} = \frac{\frac{\Delta Y}{Y}}{\Delta X} = \frac{\Delta \%}{\Delta X} = \frac{\% \Delta in Y}{\Delta in X} = \text{semielasticity of } Y \text{ w.r.t. } X$$

Summary:

Model	Dependent Variable	Independent Variable	Interpretation of β_1
Level-Level	Y	X	$\Delta Y = \beta_1 \Delta X$
Level-Log	Y	$\log X$	$\Delta Y = (\beta_1/100)\% \Delta X$
Log-Level	$\log(Y)$	X	$\% \Delta Y = (100\beta_1) \Delta X$
Log-Log	$\log(Y)$	$\log(X)$	$\% \Delta Y = \beta_1 \% \Delta X$

3 The Least Squares Estimator

The data consists of n vectors $(X_{i1}, X_{i2}, \dots, X_{ik}, Y_i)$, $i = 1, \dots, n$

The LSE (least squares estimator) $\hat{\beta}$, of β (the one from the BLP) the one that minimizes:

$$SS(\mathbf{c}) = \frac{1}{n} \sum_{i=1}^n (Y_i - c_0 - c_1 X_{i2} - \dots - c_k X_{ik})^2$$

$$\hat{\beta} = \operatorname{argmin}_{\mathbf{c} \in R^{k+1}} SS(\mathbf{c})$$

Definition 5.1 Given the data set $(X_{i1}, X_{i2}, \dots, X_{ik}, Y_i)$, $i = 1, \dots, n$ we define

the regression matrix by $X = \begin{bmatrix} 1 & X_{11} & X_{12} & \dots & X_{1k} \\ 1 & X_{21} & X_{22} & \dots & X_{2k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n1} & X_{n2} & \dots & X_{nk} \end{bmatrix}$ and the vector of

the dependent variable as $\mathbf{Y} = (Y_1, \dots, Y_n)^T$.

Size of X is $n \times (k + 1)$. X^T has size $(k + 1) \times n$

Theorem 5.1 The LSE is the solution to the following system of equations

$$X^T X \hat{\beta} = X^T \mathbf{Y} \text{ (these are called the Normal equations)}$$

If $\det(X^T X) \neq 0$, then the unique solution to the system above is

$$\hat{\beta} = (X^T X)^{-1} X^T \mathbf{Y}$$

Remark: With a single explanatory variable, $X_{n \times 2} = \begin{bmatrix} 1 & X_{11} \\ \vdots & \vdots \\ 1 & X_{n1} \end{bmatrix}$, for this

case you should check that indeed $\hat{\beta} = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix} = \begin{pmatrix} \bar{Y} - \hat{\beta}_1 \bar{X} \\ \frac{S_{xy}}{S_{xx}} \end{pmatrix}$ correspond to the formula above.

Proof: We will consider the first order conditions of $\frac{1}{n} \sum_{i=1}^n (Y_i - C_0 - C_1 X_{i1} - \dots - C_k X_{ik})^2 = SS(\mathbf{C})$ with respect to $\mathbf{C} = (C_0, C_1, \dots, C_k)^T$.

$$\frac{\partial SS(\mathbf{C})}{\partial C_0} = -\frac{2}{n} \sum_{i=1}^n (Y_i - C_0 - C_1 X_{i1} - \dots - C_k X_{ik})$$

and for $j=1, \dots, k$

$$\frac{\partial SS(\mathbf{C})}{\partial C_j} = -\frac{2}{n} \sum_{i=1}^n (Y_i - C_0 - C_1 X_{i1} - \dots - C_k X_{ik}) X_{ij}$$

Then $\widehat{\beta}$ must solve the FOC's: $\frac{\partial SS(\widehat{\beta})}{\partial C_0} = 0$ and $\frac{\partial SS(\widehat{\beta})}{\partial C_j} = 0$ for $j = 1, \dots, k$. Substituting this into the expressions above yields, after manipulates:

$$\begin{aligned} \widehat{\beta}_0 n + \widehat{\beta}_1 \sum_{i=1}^n X_{i1} + \dots + \widehat{\beta}_k \sum_{i=1}^n X_{ik} &= \sum_{i=1}^n Y_i \\ \widehat{\beta}_0 \sum_{i=1}^n X_{i1} + \widehat{\beta}_1 \sum_{i=1}^n X_{i1}^2 + \dots + \widehat{\beta}_k \sum_{i=1}^n X_{ik} &= \sum_{i=1}^n Y_i X_{i1} \\ \vdots \\ \widehat{\beta}_0 \sum_{i=1}^n X_{ik} + \widehat{\beta}_1 \sum_{i=1}^n X_{i1} X_{ik} + \dots + \widehat{\beta}_k \sum_{i=1}^n X_{ik}^2 &= \sum_{i=1}^n Y_i X_{ik} \end{aligned}$$

Writing this in matrix form:

$$\begin{bmatrix} n & \sum X_{i1} & \dots & \sum X_{ik} \\ \sum X_{i1} & \sum X_{i1}^2 & \dots & \sum X_{i1} X_{ik} \\ \sum X_{i2} & \sum X_{i2} X_{i1} & \dots & \sum X_{i2} X_{ik} \\ \vdots & \vdots & \ddots & \vdots \\ \sum X_{ik} & \sum X_{ik} X_{i1} & \dots & \sum X_{ik}^2 \end{bmatrix} \times \begin{bmatrix} \widehat{\beta}_0 \\ \widehat{\beta}_1 \\ \widehat{\beta}_2 \\ \vdots \\ \widehat{\beta}_k \end{bmatrix} = \begin{bmatrix} \sum Y_i \\ \sum Y_i X_{i1} \\ \sum Y_i X_{i2} \\ \vdots \\ \sum Y_i X_{ik} \end{bmatrix}$$

$$X^T X = \begin{bmatrix} n & \sum X_{i1} & \cdots & \sum X_{ik} \\ \sum X_{i1} & \sum X_{i1}^2 & \cdots & \sum X_{i1} X_{ik} \\ \sum X_{i2} & \sum X_{i2} X_{i1} & \cdots & \sum X_{i2} X_{ik} \\ \vdots & \vdots & \ddots & \vdots \\ \sum X_{ik} & \sum X_{ik} X_{i1} & \cdots & \sum X_{ik}^2 \end{bmatrix}, X^T Y = \begin{bmatrix} \sum Y_i \\ \sum Y_i X_{i1} \\ \sum Y_i X_{i2} \\ \vdots \\ \sum Y_i X_{ik} \end{bmatrix}$$

Therefore, the LSE is the solution to the system:

$$X^T X \hat{\beta} = X^T Y$$

If $\det(X^T X) \neq 0$ (this means, the inverse exists), then the unique solution is given by

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

Remark: $X^T X_{(k+1) \times (k+1)}$ ■

The condition that $\det(X^T X) \neq 0$ happens when $\text{rank}(X^T X) = k+1$. From Linear Algebra, $\text{rank}(X^T X) = \text{rank}(X)$.

(Note: rank=dimension of column space, i.e., the number of linearly independent columns)

For econometric application, since n (number of observations) $> k+1$ (number of explanatory variables), this rank condition is usually met, $\text{rank}(X) \leq k+1$ and if $\text{rank}(X) = k+1$ then LSE exists and is unique.

It may be shown that the Hessian of $SS(\mathbf{C})$ (matrix of second order derivatives) is $X^T X$. It is also true that if $\text{rank}(X^T X) = k+1$ than $X^T X$ is a positive definite matrix (For every vector $z \in R^{k+1}$, $z^T X^T X z \geq 0$) and in this case $\hat{\beta}$ is the unique global minimum of $SS(\mathbf{C})$, that is $SS(\mathbf{C}) > SS(\hat{\beta})$, $\forall \mathbf{C} \neq \hat{\beta}$

4 Geometry of Least Squares

Math Detour:

Consider the vector space $V = R^n$ over the field $k = R$. Linear Combination: if $v_1, \dots, v_n$ are vectors and $\lambda_1, \dots, \lambda_n$ are scalars, then the linear combination of those vectors with these scalars is

$$\lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_n v_n$$

Column space of a matrix: Let A be a $n \times m$ matrix. $A = [a_1, \dots, a_m]$, where each a_i is a vector in R^n for $i = 1, \dots, m$. The column space of A is the set of all linear combinations of its columns:

$$L(A) = \{\lambda_1 a_1 + \dots + \lambda_m a_m \mid \text{where } \lambda_1, \dots, \lambda_m \in R \text{ are scalars}\}$$

Definition 5.2 Let $\widehat{\beta}$ be the LSE, the fitted model is

$$\widehat{E}^*(Y|X_i, \dots, X_k) = \widehat{\beta}_0 + \widehat{\beta}_1 X_1 + \dots + \widehat{\beta}_k X_k$$

The fitted values are:

$$\widehat{Y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 X_{i1} + \widehat{\beta}_2 X_{i2} \dots + \widehat{\beta}_k X_{ik}$$

OR in vector form

$$\widehat{\mathbf{Y}} = (\widehat{Y}_1, \dots, \widehat{Y}_n)^T$$

The residuals are:

$$\widehat{\varepsilon}_i = Y_i - \widehat{Y}_i \text{ for } i = 1, \dots, n$$

OR in the vector form

$$\widehat{\varepsilon} = \mathbf{Y} - \widehat{\mathbf{Y}}$$

Observe that the fitted values in vector form can be expressed as follows:

$$\hat{\mathbf{Y}} = X\hat{\boldsymbol{\beta}}$$

then since $\hat{\boldsymbol{\beta}} = (X^T X)^{-1} X^T \mathbf{Y} \Rightarrow \hat{\mathbf{Y}} = X(X^T X)^{-1} X^T \mathbf{Y}$

If we define $P_X = X(X^T X)^{-1} X^T$ then $\hat{\mathbf{Y}} = P_X \mathbf{Y}$

P_X is a projection matrix : P_X projects vectors in R^n to the column space of X , i.e. P_X projects only $\mathbf{Y} \in R^n$ to $L(X)$

↓

LSE produces a vector $\hat{\mathbf{Y}}$ that minimizes the distance between $\mathbf{Y}$ to the column space of X , $L(X)$

Note: A projection matrix is a square matrix that is

- symmetric $A^T = A$
- idempotent $A^2 = A$

In a similar fashion, the vector of residuals can be expressed as follows:

$$\hat{\epsilon} = \mathbf{Y} - \hat{\mathbf{Y}} = \mathbf{Y} - P_X \mathbf{Y} = I\mathbf{Y} - P_X \mathbf{Y} = (I - P_X)\mathbf{Y}$$

$$\Rightarrow \hat{\epsilon} = (I - P_X)\mathbf{Y}$$

$$\text{If we define } N_X = (I - P_X) = I - X(X^T X)^{-1} X^T \Rightarrow \hat{\epsilon} = N_X \mathbf{Y}$$

N_X is also a projection matrix: N_X projects any vector in R^n into the orthogonal complement of $L(X)$, which is denoted $L(X)^\perp$

N_X takes the vector $\mathbf{Y}$ and produces $\hat{\epsilon} \in L(X)^\perp$

Orthogonality of two vectors, say ω and v means that their dot product is

0:

$$v \cdot \omega = \sum_{i=1}^r v_i \omega_i = 0$$

$$\text{If } v \in L(X) \text{ and } \omega \in L(X)^\perp \Rightarrow v \cdot \omega = 0$$

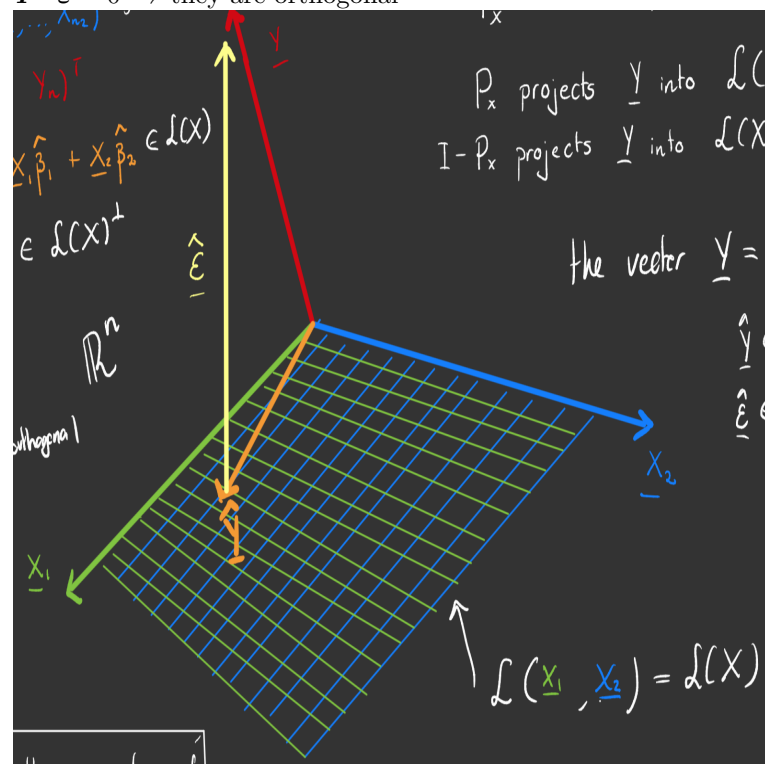
$$R^n = L(X) \oplus L(X)^\perp$$

$$\text{direct sum: } \mathbf{Y} = v + \omega, \text{ where } v \in L(X), \omega \in L(X)^\perp$$

$P_X = X(X^T X)^{-1} X^T$ is the projection matrix P_X projects $\mathbf{Y}$ into $L(X)$ to obtain $\hat{\mathbf{Y}}$. $I - P_X$ projects $\mathbf{Y}$ into $L(X)^\perp$ to obtain $\hat{\boldsymbol{\varepsilon}}$
 The vector $\mathbf{Y} = \hat{\mathbf{Y}} + \hat{\boldsymbol{\varepsilon}}$, $\hat{\mathbf{Y}} \in L(X)$, $\hat{\boldsymbol{\varepsilon}} \in L(X)^\perp$

$$\begin{aligned} \mathbf{X}_1 &= (X_{11}, X_{21}, \dots, X_{n1})^T \\ \mathbf{X}_2 &= (X_{12}, X_{22}, \dots, X_{n2})^T \hookrightarrow X = (\mathbf{X}_1, \mathbf{X}_2) \\ \mathbf{Y} &= (Y_1, \dots, Y_n)^T \\ \hat{\mathbf{Y}} &= X\hat{\boldsymbol{\beta}} = \mathbf{X}_1\hat{\beta}_1 + \mathbf{X}_2\hat{\beta}_2 \in L(X) \\ \hat{\boldsymbol{\varepsilon}} &= \mathbf{Y} - \hat{\mathbf{Y}} \in L(X)^\perp \end{aligned}$$

$\hat{\mathbf{Y}} \cdot \hat{\boldsymbol{\varepsilon}} = 0 \Rightarrow$ they are orthogonal



Theorem 5.2 Suppose X has full column rank, then:

- (i) $P_X = X(X^T X)^{-1} X^T$ is a projection matrix onto $L(X)$
- (ii) $N_X = I - P_X$ is a projection matrix onto $L(X)^\perp$
- (iii) $P_X X = X$
- (iv) $N_X X = 0$ (0 matrix)

Proof: (i) First we show symmetry:

$$\begin{aligned} P_X^T &= (X(X^T X)^{-1} X^T)^T = (X^T)^T [(X^T X)^{-1}]^T X^T \\ &= X(X^T X)^{-1} X^T \\ &= P_X \end{aligned}$$

Now idempotency:

$$\begin{aligned} P_X P_X &= (X(X^T X)^{-1} X^T)(X(X^T X)^{-1} X^T) \\ &= X(X^T X)^{-1} X^T X(X^T X)^{-1} X^T \\ &= X I (X^T X)^{-1} X^T \\ &= X(X^T X)^{-1} X^T \\ &= P_X \end{aligned}$$

Remark: I is the identity matrix of size $k+1 \times k+1$
 P_X is of size $n \times n$. $X^T X$ is of size $(k+1)(k+1)$. X is of size $n \times k+1$
Homework: finish the proof by showing (ii), (iii), (iv).

Once again we decompose

$$\begin{aligned}
\mathbf{Y} &= \hat{\mathbf{Y}} + \hat{\boldsymbol{\varepsilon}} \text{ where } \hat{\mathbf{Y}} \in L(X), \hat{\boldsymbol{\varepsilon}} \in L(X)^\perp \\
&= P_X \mathbf{Y} + N_X \mathbf{Y} \\
&= (P_X + N_X) \mathbf{Y} \\
&= (P_X + I - P_X) \mathbf{Y} \\
&= I \mathbf{Y} \\
&= \mathbf{Y}
\end{aligned}$$

$\hat{\mathbf{Y}}$ is the component explained by the BLP

$\hat{\boldsymbol{\varepsilon}}$ is the component not explained by the BLP

$\hookrightarrow$ in the sense of being explained by the model as represented by the columns
of X , i.e. the explanatory variables.

$$\hat{\mathbf{Y}} \cdot \hat{\boldsymbol{\varepsilon}} = \hat{\mathbf{Y}}^T \hat{\boldsymbol{\varepsilon}} = 0$$

$\dim(L(X)) = k + 1$, $\dim(L(X))^\perp = n - k + 1$ by the rank-nullity theorem.

5 Goodness of Fit Statistics

Math-Detour

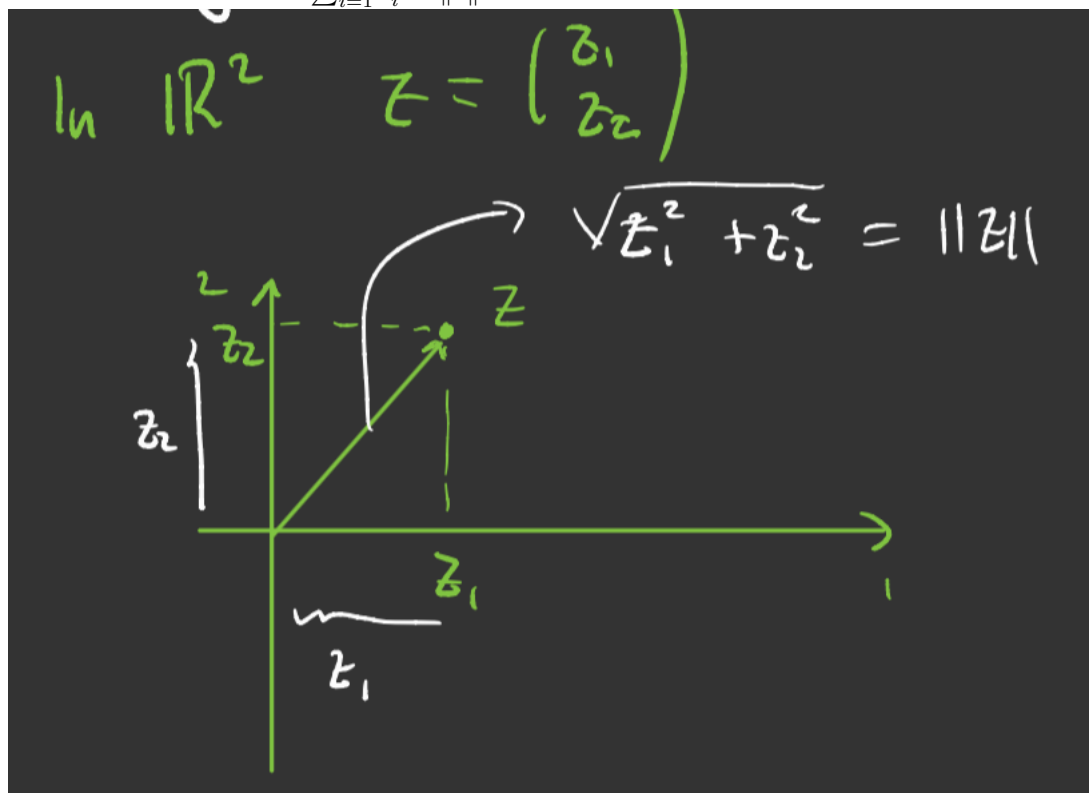
Dot product: For two vectors $z = \begin{pmatrix} z_1 \\ \vdots \\ z_n \end{pmatrix}$, $w = \begin{pmatrix} w_1 \\ \vdots \\ w_n \end{pmatrix}$, the dot product is:

$$z \cdot w = \sum_{i=1}^n z_i w_i \text{ OR alternatively } z^T w = (z_1, \dots, z_n) \begin{pmatrix} w_1 \\ \vdots \\ w_n \end{pmatrix} = \sum_{i=1}^n z_i w_i$$

Vector-Norm: The norm (or length) of a vector z is given by:

$$\|z\| = \left(\sum_{i=1}^n z_i^2 \right)^{1/2}$$

Notice that $z \cdot z = z^T z = \sum_{i=1}^n z_i^2 = \|z\|^2$



Recall we have the decomposition:

$$\mathbf{Y} = \hat{\mathbf{Y}} + \hat{\boldsymbol{\varepsilon}}$$

$$\mathbf{Y}^T \mathbf{Y} = (\hat{\mathbf{Y}} + \hat{\boldsymbol{\varepsilon}})^T (\hat{\mathbf{Y}} + \hat{\boldsymbol{\varepsilon}}) = (\hat{\mathbf{Y}}^T + \hat{\boldsymbol{\varepsilon}}^T) (\hat{\mathbf{Y}} + \hat{\boldsymbol{\varepsilon}}) = \hat{\mathbf{Y}}^T \hat{\mathbf{Y}} + \hat{\mathbf{Y}}^T \hat{\boldsymbol{\varepsilon}} + \hat{\boldsymbol{\varepsilon}}^T \hat{\mathbf{Y}} + \hat{\boldsymbol{\varepsilon}}^T \hat{\boldsymbol{\varepsilon}}$$

because of the orthogonality of $\hat{\mathbf{Y}}$ and $\hat{\boldsymbol{\varepsilon}}$ $\hat{\boldsymbol{\varepsilon}} = \hat{\mathbf{Y}}^T \hat{\mathbf{Y}} + \hat{\boldsymbol{\varepsilon}}^T \hat{\boldsymbol{\varepsilon}}$
using the notion of norm of a vector

$$\|\mathbf{Y}\|^2 = \|\hat{\mathbf{Y}}\|^2 + \|\hat{\boldsymbol{\varepsilon}}\|^2$$

If we divide the both sides by $\|\hat{\mathbf{Y}}\|^2$, we obtain

$$1 = \frac{\|\mathbf{Y}\|^2}{\|\mathbf{Y}\|^2} = \frac{\|\hat{\mathbf{Y}}\|^2}{\|\mathbf{Y}\|^2} + \frac{\|\hat{\boldsymbol{\varepsilon}}\|^2}{\|\hat{\mathbf{Y}}\|^2} \text{ OR in a percentage terms:}$$

$$100\% = \frac{\|\hat{\mathbf{Y}}\|^2}{\|\mathbf{Y}\|^2} (100\%) + \frac{\|\hat{\boldsymbol{\varepsilon}}\|^2}{\|\mathbf{Y}\|^2} (100\%)$$

Appealing to the idea that:

- $\hat{\mathbf{Y}}$ is the part of $\mathbf{Y}$ explained by X
- $\hat{\varepsilon}$ is the part of $\mathbf{Y}$ not explained by X

it is interesting to look at the following term:

$$\frac{\|\hat{\mathbf{Y}}\|^2}{\|\mathbf{Y}\|^2} 100\%$$

- Intuitively, this is the part of $\mathbf{Y}$ explained by the model (X)
- The part of the dependent variable explained by the independent variables

$\hookrightarrow$ Since this is a quotient involving sums of squares, it is related to the variability of $\mathbf{Y}$. However, since the model includes a constant, $\|\hat{\mathbf{Y}}\|^2$ and $\|\mathbf{Y}\|^2$ contain the influence of the intercept of the model. Since an intercept is a constant $\Rightarrow$ there is no variability in it $\Rightarrow$ it is inflating the quotient above

Thus, it is relevant to cancel the influence of the constant intercept.

We are going to achieve this by "detrending around the mean" each component, i.e., subtract from each component its mean.

$\hookrightarrow$ we will do this using a special projection matrix

Define $\mathbf{1} = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}$, $\mathbf{1}$ is a vector of 1's of size $n \times 1$. Consider the projection matrices attached to this vector:

$$P_{\mathbf{1}} = \mathbf{1}(\mathbf{1}^T \mathbf{1})^{-1} \mathbf{1}^T$$

$$N_{\mathbf{1}} = I - P_{\mathbf{1}}$$

, where I is the identity matrix of size $n \times n$

$P_{\mathbf{1}}$ projects into the $L(\mathbf{1}) \longrightarrow P_{\mathbf{1}}$ computes the mean of any vector

$N_{\mathbf{1}}$ projects into the $L(\mathbf{1})^{\perp} \longrightarrow N_{\mathbf{1}}$ computes the deviation from the mean of any vector

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix} \Rightarrow P_1 \mathbf{Y} = \begin{pmatrix} \bar{Y} \\ \vdots \\ \bar{Y} \end{pmatrix} \text{ where } \bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$$

$$N_1 \mathbf{Y} = \begin{bmatrix} Y_1 - \bar{Y} \\ Y_2 - \bar{Y} \\ \vdots \\ Y_n - \bar{Y} \end{bmatrix}$$

From $\mathbf{Y} = \hat{\mathbf{Y}} + \hat{\boldsymbol{\varepsilon}}$, multiply by N_1 to obtain

$$N_1 \mathbf{Y} = N_1 \hat{\mathbf{Y}} + N_1 \hat{\boldsymbol{\varepsilon}}$$

Lemma: If $\mathbf{1} \in L(X)$,

(the sample mean of $\hat{\mathbf{Y}}$) $\bar{\hat{\mathbf{Y}}} = \bar{Y}$

(the sample mean of $\hat{\boldsymbol{\varepsilon}}$) $\bar{\hat{\boldsymbol{\varepsilon}}} = 0$

Proof: Homework (will appear in a PS)

$N_1 \mathbf{Y} = N_1 \hat{\mathbf{Y}} + N_1 \hat{\boldsymbol{\varepsilon}} (*)$
 Since $\bar{\boldsymbol{\varepsilon}} = 0$, $N_1 \hat{\boldsymbol{\varepsilon}} = \hat{\boldsymbol{\varepsilon}}$, and that

$$N_1 \mathbf{Y} = \begin{pmatrix} Y_1 - \bar{Y} \\ \vdots \\ Y_n - \bar{Y} \end{pmatrix}$$

$$N_1 \hat{\mathbf{Y}} = \begin{pmatrix} \hat{Y}_1 - \hat{\bar{Y}} \\ \vdots \\ \hat{Y}_n - \hat{\bar{Y}} \end{pmatrix} = \begin{pmatrix} \hat{Y}_1 - \bar{Y} \\ \vdots \\ \hat{Y}_n - \bar{Y} \end{pmatrix}$$

It is also the case that $N_1 \hat{\mathbf{Y}}$ and $\hat{\boldsymbol{\varepsilon}}$ are orthogonal (verify by computing $(N_1 \hat{\mathbf{Y}})^T \hat{\boldsymbol{\varepsilon}}$)
 Thus we obtain from (*)

$$(N_1 \mathbf{Y})^T (N_1 \mathbf{Y}) = (N_1 \hat{\mathbf{Y}} + N_1 \hat{\boldsymbol{\varepsilon}})^T (N_1 \hat{\mathbf{Y}} + N_1 \hat{\boldsymbol{\varepsilon}})$$

$$\mathbf{Y}^T N_1 N_1 \mathbf{Y} = (\hat{\mathbf{Y}}^T N_1 + N_1 \hat{\boldsymbol{\varepsilon}}^T) (N_1 \hat{\mathbf{Y}} + N_1 \hat{\boldsymbol{\varepsilon}})$$

$$\mathbf{Y}^T N_1 \mathbf{Y} = \hat{\mathbf{Y}}^T N_1 N_1 \hat{\mathbf{Y}} + \hat{\mathbf{Y}}^T N_1 \hat{\boldsymbol{\varepsilon}} + \hat{\boldsymbol{\varepsilon}}^T N_1 \hat{\mathbf{Y}} + \hat{\boldsymbol{\varepsilon}}^T \hat{\boldsymbol{\varepsilon}}$$

$$\mathbf{Y}^T N_1 \mathbf{Y} = \hat{\mathbf{Y}}^T N_1 \hat{\mathbf{Y}} + \hat{\boldsymbol{\varepsilon}}^T \hat{\boldsymbol{\varepsilon}}$$

$$\|N_1 \mathbf{Y}\|^2 = \|N_1 \hat{\mathbf{Y}}\|^2 + \|\hat{\boldsymbol{\varepsilon}}\|^2 \quad (**)$$

$\hookrightarrow$ ANOVA Identity = Analysis of variance. Giving the decomposition of the variance of $\bar{Y}$ in explained and unexplained components.

The ANOVA identity is usually presented with the following notation:

$$\begin{aligned} SST &= \sum_{i=1}^n (Y_i - \bar{Y})^2 = \text{Sum of Squares of the total} \\ SSR &= \sum_{i=1}^n (\widehat{Y}_i - \bar{Y})^2 = \text{Sum of Squares of the regression} \\ SSE &= \sum_{i=1}^n \widehat{\varepsilon}_i^2 = \text{Sum of Squares of the error} \end{aligned}$$

Theorem 5.3 If $\mathbf{Y} \in R^n$, $X_{n \times (k+1)}$ is a regression matrix of full column rank and $\mathbf{1} \in L(X)$. Then the following identity is true:

$$(ANOVA) : SST = SSR + SSE (+)$$

Proof: Since $\mathbf{1} \in L(X)$, then from the Lemma $\widehat{\widehat{\mathbf{Y}}} = \bar{Y}$ and $\widehat{\widehat{\varepsilon}} = 0$, and also $N_1 \widehat{\mathbf{Y}}$ and $\widehat{\varepsilon}$ are orthogonal; then clearly

$$\begin{aligned} \|N_1 \mathbf{Y}\|^2 &= \|(Y_1 - \bar{Y}, \dots, Y_n - \bar{Y})^T\| = SST \\ \|N_1 \widehat{\mathbf{Y}}\|^2 &= \|(\widehat{Y}_1 - \bar{Y}, \dots, \widehat{Y}_n - \bar{Y})^T\| = SSR \end{aligned}$$

Therefore the equation (+) follows from (**) in the last page

Using the idea of decomposing the variance into the explained and unexplained components, we define the goodness of fit measure, called R^2 (R-squared) as:

$$R^2 = \frac{\|N_1 \hat{\mathbf{Y}}\|^2}{\|N_1 \mathbf{Y}\|^2} \times (100\%) = \frac{SSR}{SST} \times 100\%$$

:Intuitively is the percentage of the total variation of $\mathbf{Y}$ explained by the regression model.

- $0 \leq R^2 \leq 100\%$ (it is like a squared correlation coefficient)
- $R^2 = 100\% - \frac{SSE}{SST} \cdot 100\% = (1 - \frac{SSE}{SST})(100\%)$

$\frac{SSE}{SST} \cdot 100\%$ is the percentage of the total variation of $\mathbf{Y}$ not explained by the model.

6 The monotonicity of the R^2

The R^2 is non-decreasing on the number of explanatory variables. To see this:

$$SST = SSR(X_1, \dots, X_k) + SSE(X_1, \dots, X_k)$$

Notice that SST is the same always, irrespective of variables I'm using. $\Rightarrow$ If we add one variable X_{k+1} we have

$$SST = SSR(X_1, \dots, X_k, X_{k+1}) + SSE(X_1, \dots, X_k, X_{k+1})$$

We can show that: $SSE(X_1, \dots, X_k, X_{k+1}) \leq SSE(X_1, \dots, X_k) \Rightarrow SSR(X_1, \dots, X_k, X_{k+1}) \geq SSR(X_1, \dots, X_k)$ (One proof of this will be asked in a Problem Set)

$L(X_1, \dots, X_k) \subseteq L(X_1, \dots, X_k, X_{k+1}) \Rightarrow \hat{\mathbf{Y}}(X_1, \dots, X_k, X_{k+1})$ will fit no worse than $\hat{\mathbf{Y}}(X_1, \dots, X_k)$

$\Rightarrow$ The sum of the squares of two errors of the regression with more variables will be not larger than one from the shorter one.

$\therefore$ expressing the R^2 as function of the regression:

$$R^2(X_1, \dots, X_k) = \frac{SSR(X_1, \dots, X_k)}{SST}$$

$\Rightarrow$ we have the monotonicity of the R^2 :

$$R^2(X_1, \dots, X_k, X_{k+1}) = \frac{SSR(X_1, \dots, X_k, X_{k+1})}{SST} \geq \frac{SSR(X_1, \dots, X_k)}{SST} = R^2(X_1, \dots, X_k)$$

This inequality hold no matter what is in X_{k+1}

$\hookrightarrow$ Because of this reason other goodness-of-fit statistics have been invented that penalize the fact adding explanatory variables with little-to-none explanatory power.

Using these ideas, we define the "adjusted R-squared"

$$\bar{R}^2 = (1 - \frac{\hat{\sigma}^2}{\hat{\sigma}_y^2}) \times 100\%$$

where $\hat{\sigma}_y^2 = \frac{SST}{n-1}$, $\hat{\sigma}^2 = \frac{SSE}{n-(k+1)}$

$\hat{\sigma}_y^2$ is an estimate of the variance of Y

$\hat{\sigma}^2$ is like an estimate of the conditional variance of $Y|X$ (like in Lecture 3) $\hookrightarrow$
this is an intuitive explanation.

- $\bar{R}^2$ is not monotonic with respect to the number of explanatory variables used.
- There is a correlation between the R^2 and $\bar{R}^2$ given by the following formula

$$1 - \bar{R}^2 = 1 - (1 - \frac{\hat{\sigma}^2}{\hat{\sigma}_y^2}) = \frac{\hat{\sigma}^2}{\hat{\sigma}_y^2} = \frac{SSE/n - k - 1}{SST/n - 1} = \frac{SSE}{SST} (\frac{n - 1}{n - k - 1}) = (1 - R^2) (\frac{n - 1}{n - k - 1})$$

$$\therefore 1 - \bar{R}^2 = \frac{n-1}{n-k-1}(1 - R^2)$$

Example: n=100, we have two models

Model	k	R^2	$1 - \bar{R}^2$	$\bar{R}^2$
1	4	0.85	0.1633	0.837
2	7	0.851	0.1738	0.826

From 4 to 7 explanatory variables (add 3)

$$R^2_{model1} = 85\% \longrightarrow R^2_{model2} = 85.1\%$$

$$\bar{R}^2_{model1} = 83.7\% \longrightarrow \bar{R}^2_{model2} = 82.6\%$$

This is indicative of the poor explanatory power of those extra 3 variables.

7 Consistency of the LSE

Data Generating Process (DGP): process which generates the data.

· will satisfy certain assumptions

↔ For example, in Lecture 3, the DGP satisfied assumptions A1-A5 which are called the classical linear regression assumptions.

↔ LSE was shown to be conditionally unbiased and had conditional normal distribution

↔ These assumptions are problematic

A2: Linearity of the CEF

A4: $Var(Y|X) = \sigma^2$ (homoskedasticity)

A5: $Y|X \sim Normal$

↔ These assumptions require complete knowledge of the distribution and being able to assert that the CEF is linear, the conditional variance is constant, and normality of $Y|X$, **which is usually violated in applications.**

The goal is to relax some of this assumptions and show that $\hat{\beta} \xrightarrow{p} \beta$

We will assume that the DGP satisfies:

Assumption 1M (A1M): For every sample size n , the data is a random sample of independent and identically distributed vectors:

$(X_{11}, \dots, X_{1k}, Y_1), \dots, (X_{n1}, \dots, X_{nk}, Y_n)$. Y is the dependent variable and $X_2, \dots, X_k$ are the explanatory variable. All variables have finite variances and $\det(X(\mathbf{X})) > 0$

Note $\mathbf{X} = \begin{pmatrix} X_1 \\ \vdots \\ X_k \end{pmatrix}$; X_{ij} are realizations of the random variable X_j for $i = 1, \dots, n$, $j = 1, \dots, k$

Assumption 2M (A2M): The regression matrix $X = \begin{bmatrix} 1 & X_{11} & X_{12} & \cdots & X_{1k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n1} & X_{n2} & \cdots & X_{nk} \end{bmatrix}$

is of full column rank for every n .

A set of vectors $v_1, \dots, v_k$ is linearly independent if the only numbers $\lambda_1, \dots, \lambda_k$ such that $\lambda_1 v_1 + \dots + \lambda_k v_k = 0$ are $\lambda_1 = \lambda_2 = \dots = \lambda_k = 0$

When $k = 1$, A2M is equivalent to $S_{xx} = \sum_{i=1}^n (X_i - \bar{X})^2 > 0$ which was assumption A3 from Lecture 3.

In simple linear regression; the BLP is $E^*(Y|X) = \beta_0 + \beta_1 X$ where $\beta_0 = E(Y) - \beta_1 E(X)$ and $\beta_1 = \frac{Cov(Y,X)}{Var(X)}$

The LSE is $\hat{\beta}_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{S_{xy}}{S_{xx}}$ and $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$

Consistency of $\hat{\beta}_1$: Write $\hat{\beta}_1 = \frac{\frac{1}{n} S_{xy}}{\frac{1}{n} S_{xx}}$, from the results from Lecture 4

$$\begin{aligned} \frac{1}{n} S_{xy} &\xrightarrow{p} Cov(X, Y) \\ \frac{1}{n} S_{xx} &\xrightarrow{p} Var(X) \end{aligned}$$

and from Theorem 4.3 (Slutsky's Theorem) the ratio converges in probability to $\frac{Cov(X,Y)}{Var(X)} = \beta_1 \therefore \hat{\beta}_1 \xrightarrow{p} \beta_1$

Consistency of $\hat{\beta}_0$: Note by the WLLN, $\bar{Y} \xrightarrow{p} \mu_Y$, $\bar{X} \xrightarrow{p} \mu_X$ and since $\hat{\beta}_1 \xrightarrow{p} \beta_1$ by Theorem 4.3, $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X} \xrightarrow{p} \mu_Y - \beta_1 \mu_X = E(Y) - \beta_1 E(X) = \beta_0 \therefore \hat{\beta}_0 \xrightarrow{p} \beta_0$

We generalize the discussion to k explanatory variables.
Recall the BLP with k explanatory variables (Theorem 2.10)

$$E^*(Y|X_1, \dots, X_k) = \beta_0 + \beta_1 X_1 + \dots + \beta_k X_k$$

where

$$\beta_0 = E(Y) - \beta_1 E(X_1) - \dots - \beta_k E(X_k)$$

$$\boldsymbol{\beta}_p = \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_k \end{pmatrix} = (V(\mathbf{X}))^{-1} Cov(\mathbf{X}, Y)$$

Theorem 5.4 Let $\hat{\boldsymbol{\beta}}$ be the LSE of $\boldsymbol{\beta}$ (the vector of coefficients of the BLP). Then under assumptions A1M & A2M

$$\hat{\boldsymbol{\beta}} \xrightarrow[p]{} \boldsymbol{\beta} \text{ as } n \rightarrow \infty \text{ i.e. } \hat{\boldsymbol{\beta}} \text{ is consistent}$$

Proof: The case $k = 2$ is general enough. We have that

$$X = \begin{pmatrix} 1 & X_{11} & X_{12} \\ 1 & X_{21} & X_{22} \\ \vdots & \vdots & \vdots \\ 1 & X_{n1} & X_{n2} \end{pmatrix} \text{ and } \mathbf{Y} = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix}$$

$$\text{with } \hat{\beta} = (X^T X)^{-1} X^T Y = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix}$$

Using the notation of Lemma 3.1, consider

$$S_{x_i, x_j} = \sum_{l=1}^n (X_{li} - \bar{X}_i)(X_{lj} - \bar{X}_j)$$

$$S_{x_i, y} = \sum_{l=1}^n (X_{li} - \bar{X}_i)(Y_l - \bar{Y})$$

$$S_{\mathbf{x}\mathbf{x}} = \begin{bmatrix} S_{x_1, x_1} & S_{x_1, x_2} \\ S_{x_2, x_1} & S_{x_2, x_2} \end{bmatrix} \text{ and } S_{\mathbf{x}, y} = \begin{pmatrix} S_{x_1, y} \\ S_{x_2, y} \end{pmatrix}$$

(Homework and PS) It can be shown that:

$$\begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix} = (S_{\mathbf{x}\mathbf{x}})^{-1} S_{\mathbf{x}y} = \left(\frac{1}{n} S_{\mathbf{x}\mathbf{x}}\right)^{-1} \left(\frac{1}{n}\right) S_{\mathbf{x}y}$$

Now, $\frac{1}{n} S_{\mathbf{x}\mathbf{x}} \xrightarrow{p} V(\mathbf{X})$ and $\left(\frac{1}{n}\right) S_{\mathbf{x}y} \xrightarrow{p} \begin{pmatrix} Cov(X_1, Y) \\ Cov(X_2, Y) \end{pmatrix}$ **applications of the WLLN**

Since matrix inversion is a continuous operation when the determinant of the matrix is different from zero, and $det(V(\mathbf{X})) > 0$ by A1M then from Theorem 4.3 (Slutsky's)

$$\begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix} \xrightarrow{p} [V(\mathbf{X})]^{-1} \begin{pmatrix} Cov(X_1, Y) \\ Cov(X_2, Y) \end{pmatrix} = \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix}$$

To show $\hat{\beta}_0$ converges to β_0 consider:

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}_1 - \hat{\beta}_2 \bar{X}_2 \xrightarrow{p} E(Y) - \beta_1 E(X_1) - \beta_2 E(X_2) = \beta_0$$

■